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Project Title:	Mechanisms of Stochastic Resonance on Bone Cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
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	Long-duration space flight induces bone loss. This loss could lead to a significant increase in risk of fracture. A specified form of mechanical stimulation (exercise), a combination of high-frequency vibration and low-frequency mechanical loading, may be an effective countermeasure to bone loss. This phenomenon is called stochastic resonance, in which noise vibration enhances the response of a nonlinear system to a weak signal. The global goal of this proposed study is to investigate mechanisms of stochastic resonance on bone cells. Understanding the mechanisms of stochastic resonance on bone cells will provide new information about increasing bone's response to mechanical stimulation. The hypothesis of the project is that stochastic resonance acts through L-type voltage sensitive calcium channels in the cell membrane to enhance bone response to mechanical loading.		

Task Description:	We will determine whether stochastic resonance increases mechanically sensitive genes and protein products through L-VSCCs in osteoblastic (MC3T3-E1) cells. We will study the effect of mechanical loading and stochastic resonance on mice that have an ablation of a specific calcium channel subunit (a subunit of L-VSCC) that is expressed in osteoblasts. We will study which mechanically sensitive genes are indirectly influenced by L-VSCCs during stochastic resonance exercise. Understanding the mechanisms of stochastic resonance on bone cells will provide new information about increasing bone's response to mechanical stimulation. The application of stochastic resonance offers potentially a new way to treat osteoporosis and prevent bone loss in space flight in the future. In addition, identification of the biochemical pathway(s) by which mechanical loading is transduced into gene expression may help to identify new drug targets for osteoporosis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space flight induces a suppression of bone formation in both cortical, cancellous bone and a reduction of bone mass. Periosteal bone growth decreases due to microgravity. Since bone mechanical properties are determined by both bone mass and geometry, loss of bone mass and reduced periosteal apposition resulting from space flight may lead to increased bone fragility and result in higher risk of fracture. Also, the changes in bone structure under microgravity mimic such clinical situations as skeletal unloading, prolonged bedrest, paralysis, cast immobilization. The related research can also benefit to increase our knowledge of the treatment of disuse or other osteoporosis.
Task Progress:	The research team continued to study effects of stochastic resonance on bone cells. These studies aimed to explore the role of calcium channels in osteoblastic response to stochastic resonance. MC3T3-E1 osteoblastic cells were exposed to the following conditions: (1) Sine – an oscillating sinusoidal fluid flow at 2 Hz (a peak shear stress of 12 dynes/cm2), (2) Sine+Vibration – an oscillating sinusoidal fluid flow at 2 Hz of a decreased amplitude (a peak shear stress of 10 dynes/cm2) in combination with broad frequency vibrational components of 0-50 Hz (1.2 dynes/cm2 of the mean amplitude) to invoke stochastic resonance, or (3) Control – no exposure to fluid flow stimulation. The flow signals was generated using a home made oscillatory fluid flow system. Exposure to sine + vibration fluid shear elicited a significantly larger increase in activation of ERK1/2 and release of prostaglandins E2 (PGE2) over sine wave fluid shear alone. However, blocking the mechanosensitive cation-selective channels with administration of gadolinium inhibited the activational effect of vibration on ERK1/2 and PGE2 production. Inhibiting intracellular Ca2+ release by administering thapsigardin also significantly decreased the activation of ERK1/2 and PGE2 production induced by vibration. Furthermore, inhibition of the entry of calcium through the L-type voltage sensitive calcium channels with nifedipine significantly attenuated the osteoblastic response to both the sine wave and sine+vibration fluid shear stress. These data suggest that osteoblast sensitivity to stochastic resonance is mediated by Ca2+. In addition, animal studies are carried out to investigate if L-type calcium channels mediate mechanical loading or stochastic resonance induced bone adaptation using mice with knockout (KO) of Cav1.2 and Cav1.3 genes, subunits of L-type voltage sensitive calcium channel. Some preliminary data with small number of animals showed that Cav1.3 KO mice have a slightly lower bone mineral density than the wild type control. Inactivation of Cav1.3 did not abolish mechanically induced bone formation completely. Mechanical loading of the skeleton is necessary to maintain bone structure and strength. Large amplitude strains associated with vigorous activity typically result in the greatest osteogenic response; however, data suggest that low-amplitude, broad-frequency vibration results in new bone formation and may enhance adaptation through a stochastic resonance (SR) phenomenon. That is, random noise may maximally enhance bone formation to a known osteogenic stimulus. The aims of this study were to (1) assess the ability of different vibration signals to enhance cortical bone formation during short- and long-term loading and (2) determine whether vibration could affect SR in bone. Two studies were completed wherein several osteogenic loading waveforms, with or without an additive low-amplitude, broad-frequency (0-50 Hz) vibration signal were applied to the mouse ulna in axial compression. In study 1, mice were loaded short-term (30 sec/day, 2 days) with either a carrier signal alone (1 or 2 N sine waveform), vibration signal alone [0.1 N or 0.3 N root mean square (RMS)] or combined carrier and vibration signal. In study 2, mice were loaded long-term (30 sec/day, 3 days/wk, 4 wks) with a carrier signal alone (static or sine waveform), vibration signal alone (0.02 N, 0.04 N, 0.08 N or 0.25 N RMS) or combined carrier and vibration signal. Sequential calcein bone labels were administered at 2 and 4 days and at 4 and 29 days after the first day of loading in study 1 and 2, respectively; bone formation parameters and changes in geometry were measured. Combined application of the carrier and vibration signals in study 1 resulted in significantly greater bone formation than with either signal alone ($p<0.001$); however, this increase was independently explained by increased strain levels associated with additive vibration. When load and strain levels were similar across loading groups in study 2, cortical bone formation and changes in geometry were not significantly altered by vibration. These data show that low-amplitude, broad-frequency vibration superimposed onto an osteogenic waveform or vibration alone does not enhance cortical bone adaptation at the frequencies, amplitudes and loading periods tested. There is evidence that long-term loading with passive vibration signals can result in significant increases in trabecular bone mineral density and bone volume, but there is little data to support the same effect in cortical bone. Thus, vibration alone may be more effective in promoting new trabecular rather than cortical bone formation. More long-term studies are needed to determine whether vibration at different magnitudes and frequencies than those tested in the current study, alone or superimposed onto an osteogenic waveform, can lead to or enhance cortical bone formation.
Bibliography Type:	Description: (Last Updated: 12/31/2007)
Abstracts for Journals and Proceedings	Li J, Stockwell DW, Simmons CR. "Activation of Extracellular-Signal Regulated Kinase (ERK1/2) by stochastic resonance is Ca2+ dependent in MC3T3-E1 osteoblasts." Experimental Biology 2006, April 2006. Experimental Biology 2006, Abstract #250.4, April 2006. , Apr-2006
Abstracts for Journals and Proceedings	Castillo AB, Levenda JP, Pavlik M, Warden SJ, Turner CH. "The Effects of Low Amplitude, Broad Frequency Vibration on Cortical Bone Adaptation in Adult Female Mice." The 52nd Annual Meeting of Orthopedic Research Society, Chicago, IL, March 19-22, 2006. 52nd Annual Meeting of Orthopedic Research Society, Abstract # 1623, March 2006. , Mar-2006